$$\sum_{L=1}^{ORDP1} B[L] \underbrace{\frac{d^{L-1}}{dt^{L-1}} y}_{y^{(L-1)}} = N\left(y, \dots, \underbrace{\frac{d^{ORD-1}}{dt^{ORD-1}} y}_{y^{(ORD-1)}}\right)$$

*

$$y^{(L-1)} = \frac{d}{dt} y^{(L-2)}$$

$$W_\gamma^{(L-1)} = \left(DW^{(L-2)} \cdot F\right)\Big|_\gamma = s\, W_\gamma^{(L-2)} + Y^{(L-2)} F_\gamma + \mathcal{Z}_\gamma^{(L-2)}$$

$|\gamma| \geq 2$ $\qquad$ $\lambda \cdot \gamma$

$$\mathcal{Z}_\gamma = \sum_{i=1}^{NVAR} \sum_{\substack{\beta < \gamma \\ \beta + e_i}} W_{\gamma + e_i - \beta} (\gamma_i - \beta_i + 1) F_{\beta, i}$$

$$W_\gamma^{(0)}$$

$$Y = [Y^{(0)}, Y^{(1)}, \dots, Y^{(ORD-1)}]$$

$$Y^{(0)} = Y[1], \quad Y^{(ORD-1)} = Y[ORD]$$

$$W_\gamma^{(1)} = s\, W_\gamma^{(0)} + Y^{(0)} F_\gamma + \mathcal{Z}_\gamma^{(0)}$$

$$W_\gamma^{(2)} = s\, W_\gamma^{(1)} + Y^{(1)} F_\gamma + \mathcal{Z}_\gamma^{(1)} =$$

$$s^2 W_\gamma^{(0)} + (\underbrace{Y^{(1)}}_{Y[2]} + \underbrace{Y^{(0)}}_{Y[1]} s) F_\gamma + (\underbrace{\mathcal{Z}_\gamma^{(1)}}_{\mathcal{Z}_\gamma[2]} + \underbrace{\mathcal{Z}_\gamma^{(0)}}_{\mathcal{Z}_\gamma[1]} s)$$

$\vdots$

$$W_\gamma^{(L-1)} = s^{L-1} W_\gamma^{(0)} + \sum_{j=1}^{L-1} Y[L-j]\, s^{j-1} F_\gamma + \sum_{j=1}^{L-1} \mathcal{Z}_\gamma[L-j]\, s^{j-1}$$

*

$$\sum_{L=1}^{ORDP1} B[L]\, s^{L-1} W_\gamma^{(0)} +$$

switch order of summation

$$\sum_{L=2}^{ORDP1} \sum_{j=1}^{L-1} (\cdots) = \sum_{j=1}^{ORD} \sum_{L=j+1}^{ORDP1} (\cdots)$$

$$\sum_{L=\cancel{1}2}^{ORDP1} \sum_{j=1}^{L-1} B[L]\, Y[L-j]\, s^{j-1} F_\gamma +$$

$$\sum_{L=\cancel{1}2}^{ORDP1} \sum_{j=1}^{L-1} B[L]\, \mathcal{Z}_\gamma[L-j]\, s^{j-1} = \sum_{a=1}^{ORD} \sum_{L=a+1}^{ORDP1} B[L]\, s^{L-(a+1)}\, \mathcal{Z}_\gamma[a]$$

$$\left[B[1] + s\left(\overbrace{B[2]^{*} + s\left(\underbrace{B[3] + \cdots^{\Phi}}_{ORDP1\ Terms}\right)}^{\text{Horner's Method}}\right)\right] W_\gamma^{(0)} + \qquad \left[B \quad F\right]\begin{bmatrix} W_\gamma^{(0)} \\ \cancel{W_\gamma^{(ext)}}\ 0 \end{bmatrix} = B\, W_\gamma^{(0)}$$

PRECOMPUTABLE!

$$\left[D[1] + s\left(\underbrace{D[2] + \cdots}_{ORD\ Terms}\right)\right] F_\gamma + \longrightarrow \left(\sum_{j=1}^{ORD} D[j]\, s^{j-1}\right) F_\gamma$$

** where $D[j] = \sum_{L=j+1}^{ORDP1} B[L]\, Y[L-j]$

$$\left[\underbrace{\left(B[2] + s(B[3] + \cdots)\right)}_{\substack{ORD\ terms \\ \star \\ \text{recycle computation}}} \mathcal{Z}_\gamma[1] + \underbrace{\left(B[3] + \cdots\right)}_{\substack{ORD-1\ terms \\ \Phi \\ \text{recycle computation}}} \mathcal{Z}_\gamma[2] + \cdots\right]_{ORD\ Terms}$$

**

$$Y[a] = Y[1]\, \Lambda^{a-1}$$

$$D[j] = \sum_{L=j+1}^{ORDP1} B[L]\, Y[L-j] = \sum_{L=j+1}^{ORDP1} B[L]\, Y[1]\, \Lambda^{L-(j+1)}$$

$j = 1, \dots, ORD$ $\qquad$ $\Lambda^{L-(j+1)} = \Lambda^{L-(j+2)}\, \Lambda$

$$D[j+1] = \sum_{L=j+2}^{ORDP1} B[L]\, Y[1]\, \Lambda^{L-(j+2)}$$

$$\Rightarrow \begin{cases} D[ORD] = B[ORDP1]\, Y[1] \\ D[j] = D[j+1]\, \Lambda + B[j+1]\, Y[1], \quad j = 1, \dots, ORD-1 \end{cases}$$

$$\sum_{j=1}^{ORD} D[j]\, s^{j-1} \begin{bmatrix} F_{\gamma,R} \\ \cancel{F_{\gamma,\beta}}\ 0 \\ F_{\gamma,ext} \end{bmatrix} = \underbrace{\left(\sum_{j=1}^{ORD} D_R[j]\, s^{j-1}\right)}_{\text{Horner's}} F_{\gamma,R} + \underbrace{\sum_{j=1}^{ORD} \left(D_{ext}[j]\, F_{\gamma,ext}\right) s^{j-1}}_{\text{Horner's}}$$

$$C_r(s) = \sum_{j=1}^{ORD} D_r[j]\, s^{j-1}$$

resonant mode r